

Program3 : Code a dockerized Python Flask or Node.js Application

Project structure

```
|
|
|-----app.py
|
|-----requirements.txt
|
|-----Dockerfile
```

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ mkdir Program3
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ touch app.py
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ touch Dockerfile
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ touch requirements.txt
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ open app.py
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ open Dockerfile
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ open requirements.txt
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ open requirements.txt
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ cat app.py
from flask import Flask
```

```
app=Flask(__name__)
```

```
@app.route("/")
def hello():
    return "Hello, Docker!"
```

```
if __name__=="__main__":
    app.run(host="0.0.0.0",port=5000)
```

```
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ cat requirements.txt
Flask==2.3.3
```

```
app.run(host="0.0.0.0",port=5000)
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ cat Dockerfile
#Use official Python image
FROM python:latest

#Set working directory
WORKDIR /app

COPY requirements.txt .
RUN pip install --no-cache-dir -r requirements.txt

COPY . .

EXPOSE 5000

CMD ["python","app.py"]
```

Build the docker :

```
docker build -t program3
```

Run the docker :

```
docker run -it -p 5000:5000 program3
```

```

1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ docker build -t pro3 .
[+] Building 4.9s (10/10) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 247B                             0.0s
=> [internal] load metadata for docker.io/library/python:latest 4.4s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                   0.0s
=> [1/5] FROM docker.io/library/python:latest@sha256:1ad1a43b5e2478e6205 0.0s
=> [internal] load build context                                0.2s
=> => transferring context: 2.71MB                             0.1s
=> CACHED [2/5] WORKDIR /app                                    0.0s
=> CACHED [3/5] COPY requirements.txt .                          0.0s
=> CACHED [4/5] RUN pip install --no-cache-dir -r requirements.txt 0.0s
=> [5/5] COPY . .                                              0.1s
=> exporting to image                                           0.1s
=> => exporting layers                                          0.1s
=> => writing image sha256:09cc683d8c1ee5da4d5dab710774c04cf40a6bc33e0cc 0.0s
=> => naming to docker.io/library/pro3                          0.0s
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ docker run -d -p 5000:5000 --name pro3 pro3
e18daed30b801de0cda630cca20a9ac456321f69e3bf006ca86535ff7802ccc2
1rv24mc031_darshan_hegde@darshu-Inspiron-15-3520:~/Desktop/DevLab$ docker ps -a

```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
e18daed30b80	pro3	"python app.py"	4 seconds ago	Up 4 seconds	0.0.0.0:5000->5000/tcp, [::]:5000->5000/tcp	pro3

Visit localhost:5000 in browser to see output



